TRACK P (PLANETARY): BLUEPRINT

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Abstract. A proof plan, step by step, for the three frozen statements of track P: the properly degenerate KAM theorem of Chierchia–Pinzari (specialized as frozen), the analytic planar two-planet theorem with a moving Sun, and the fixed-Sun n-planet variant. Part 0 fixes conventions, restates every frozen definition and the three frozen theorems; the remaining parts are listed in `PLAN.md`. Results of track K's blueprint are cited as [K:`label`].

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1. Conventions, definitions and the frozen statements

This part fixes notation and restates, in mathematical form, the three frozen theorems of track P (owner approval D20, 27 October 2026) together with every frozen definition they use: the 33 definitions of `KAM/Frozen/Planetary/Defs/` and the three track-K definitions `KAM.IsDiophantine`, `KAM.hamiltonianVectorField`, `KAM.symplecticForm`. The Lean text is authoritative and the statements below transcribe it, with the same quantifier order and the same order in which constants are chosen. The D22 additions (P-EXP-SAFE, P-AMD-CONFINEMENT, P-SECULAR-BOX) are not frozen and are deliberately absent; a later part is reserved for them (`PLAN.md`, Part 14).

1.1. Conventions.

- *Ambient spaces.* All coordinate spaces are $\mathbb{R}^d = (\mathtt{Fin}\, d \to \mathbb{R})$ with the max norm $|v|_\infty = \max_i |v_i|$ and Lebesgue measure; $\mathrm{Ph}(d) = \mathbb{R}^d \times \mathbb{R}^d$ carries the max norm of the two components and the product Lebesgue measure `volume`. Fourier indices are $k \in \mathbb{Z}^d$. The Lean block conventions are: for $v \in \mathbb{R}^{a+b}$, the *fast* part is $v^{\mathrm{f}} = (v_i)_{i<a}$ and the *slow* part is $v^{\mathrm{s}} = (v_{a+j})_{j<b}$.
- *Phase order.* A phase point is $z = (z_1, z_2)$ with positions (angles) z_1 and momenta (actions) z_2. In the properly degenerate setting $z = (\varphi, q; I, p)$, that is $z_1 = (\varphi, q)$ and $z_2 = (I, p)$, with $\varphi \in \mathbb{R}^a$, $q \in \mathbb{R}^b$ and $I \in \mathbb{R}^a$, $p \in \mathbb{R}^b$ (`assemble`).
- *Physical distance.* On $\mathbb{R}^2$ the physical length is $|x|_2 = \sqrt{x_1^2 + x_2^2}$, never the max norm; $x \cdot y = x_1 y_1 + x_2 y_2$. Neighbourhoods in phase space and the norms $|\cdot|$ of tori data use the max norm; physical formulas use $|\cdot|_2$.
- *Bodies.* Positions of n planets are $q \in \mathbb{R}^{2n}$; planet i has position $q^{(i)} = (q_{(i,1)}, q_{(i,2)}) \in \mathbb{R}^2$, where the index of the pair $(i, k) \in \mathtt{Fin}\, n \times \mathtt{Fin}\, 2$ is $k + 2i$ (Lean `finProdFinEquiv`).

- *Torus parameters have period one* for every torus predicate below. For the cylinder tori of P-G (Definition 1.15) the *canonical fast angles keep period* 2π (CP10 footnote 1, p. 546), so the parameterization winds by 2π in the fast directions. Consequently the frequency vector w of a torus is the physical angular frequency divided by 2π (it changes frequency constants, not the dynamics or the measure ratio; `statement.md`, "Conventions"; owner question Q-P2). For the physical tori the parameter lattice is $\mathbb{Z}^d$ and the angles $2\pi\phi$ enter only through `circularPoint`.
- *Constants and quantifier order.* In each theorem, all data precede the small parameters; the proofs must keep the order of Theorems 1.28–1.30. In particular the thresholds r_*, C, c of P-G depend on the whole fixed family $(V, W, \dots, P, \dots)$, and μ_0, d, D (resp. ε_0, d, D) are chosen before μ (resp. ε).

1.2. **Track-K definitions used by track P.** The three frozen track-P theorems import exactly these frozen track-K definitions; no definition of track I or N is used.

Definition 1.1 (Diophantine vectors)**.** For $\gamma, \tau \in \mathbb{R}$ and $\omega \in \mathbb{R}^d$, ω is (γ, τ)-*Diophantine* if $\gamma/|k|_1^\tau \le |k \cdot \omega|$ for every $k \in \mathbb{Z}^d \setminus \{0\}$, where $|k|_1 = \sum_i |k_i|$ and $k \cdot \omega = \sum_i k_i \omega_i$.
Frozen Lean declaration: `KAM.IsDiophantine`
Source: HL19 (3.1), p. 14; Pos01 (1), p. 5; same as [K:`bp:p0:diophantine`].

Definition 1.2 (Hamiltonian vector field and symplectic form)**.** For $H : \mathrm{Ph}(d) \to \mathbb{R}$ and $z = (z_1, z_2)$,

$$X_H(z) = \big(\partial_{z_2} H(z),\, -\partial_{z_1} H(z)\big),$$

that is $X_H(z)_{1,i} = DH(z)(0, e_i)$ and $X_H(z)_{2,i} = -DH(z)(e_i, 0)$, with D the Fréchet derivative (`fderiv`). For $v = (v_1, v_2)$, $w = (w_1, w_2) \in \mathrm{Ph}(d)$,

$$\Omega(v, w) = \sum_i \big(v_{2,i} w_{1,i} - v_{1,i} w_{2,i}\big).$$

With z_1 the position and z_2 the momentum, $\dot z_1 = \partial_{z_2} H$, $\dot z_2 = -\partial_{z_1} H$ are the usual Hamilton equations, and $\Omega = \sum dz_{2,i} \wedge dz_{1,i}$. The condition $\Omega(Dv, Dw) = 0$ defining a Lagrangian torus is insensitive to the sign of Ω.
Frozen Lean declaration: `KAM.hamiltonianVectorField, KAM.symplecticForm`
Source: HL19 Rem. 2.1, p. 6; same as [K:`bp:p0:hamiltonian`].

1.3. **Elementary planar and index definitions.**

Definition 1.3 (Phase space)**.** $\mathrm{Ph}(d) = \mathbb{R}^d \times \mathbb{R}^d$ (canonical coordinates *(position, momentum)*).
Frozen Lean declaration: `KAM.Planetary.Phase`

Definition 1.4 (Planar length and scalar product)**.** For $v, w \in \mathbb{R}^2$: $\mathrm{sq}(v) = \sum_k v_k^2$, $|v|_2 = \sqrt{\mathrm{sq}(v)}$ and $v \cdot w = \sum_k v_k w_k$.
Frozen Lean declaration: `KAM.Planetary.planeSq, KAM.Planetary.planeLength, KAM.Planetary.planeDot`
Source: Fej04r (28), pp. 46–47 (physical Euclidean length; independent of the ambient max norm).

Definition 1.5 (Body vectors)**.** For $q \in \mathbb{R}^{2n}$ and $i < n$, $\mathrm{body}(q, i) = q^{(i)} = (q_{(i,1)}, q_{(i,2)}) \in \mathbb{R}^2$.
Frozen Lean declaration: `KAM.Planetary.bodyVector`

Definition 1.6 (Fast and slow blocks, assembly)**.** For $v \in \mathbb{R}^{a+b}$, $v^{\mathrm{f}} \in \mathbb{R}^a$ and $v^{\mathrm{s}} \in \mathbb{R}^b$ are the first a and last b coordinates. For $I, \varphi \in \mathbb{R}^a$ and $p, q \in \mathbb{R}^b$, $\mathrm{assemble}(I, \varphi, p, q) = \big((\varphi, q), (I, p)\big) \in \mathrm{Ph}(a+b)$, with the concatenated coordinate vectors.
Frozen Lean declaration: `KAM.Planetary.fast, KAM.Planetary.slow, KAM.Planetary.assemble`
Source: CP10 (1), p. 546; canonical order $(\varphi, q; I, p)$.

Definition 1.7 (Oscillator actions)**.** For $p, q \in \mathbb{R}^b$ and $j \le b$: $R_j(p,q) = (p_j^2 + q_j^2)/2$.
Frozen Lean declaration: `KAM.Planetary.oscillatorAction`
Source: CP10 (A2") and CP11 (A_2), pp. 547 and 51.

Definition 1.8 (Fast shift)**.** For $k \in \mathbb{Z}^a$, $\mathrm{shift}(k) = \big(((2\pi k_i)_{i<a}, 0_b),\, 0\big) \in \mathrm{Ph}(a+b)$: the fast angles are translated by $2\pi k$, the slow position coordinates and all momenta are unchanged.
Frozen Lean declaration: `KAM.Planetary.fastShift`

Definition 1.9 (Fast frequency map)**.** For $H_0 : \mathbb{R}^a \to \mathbb{R}$ and $I \in \mathbb{R}^a$, $\omega_{H_0}(I)_i = DH_0(I)(e_i)$ (the Fréchet derivative is 0 where H_0 is not differentiable).
Frozen Lean declaration: `KAM.Planetary.fastFrequency`
Source: CP10 (A1), p. 546.

1.4. **Definitions for the properly degenerate theorem.**

Definition 1.10 (Cylinder and its fundamental cell)**.** For $V \subset \mathbb{R}^a$ and $r \in \mathbb{R}$,

$$\mathcal{C}(V,r) = \Big\{ z \in \mathrm{Ph}(a+b) :\ z_2^{\mathrm{f}} \in V,\ \ (z_2^{\mathrm{s}})_j^2 + (z_1^{\mathrm{s}})_j^2 < r^2 \ \ \forall j \le b \Big\},$$

that is $I \in V$ and $p_j^2 + q_j^2 < r^2$ for every j: a *product of discs*, not a box. Its fundamental cell for the fast angles of period 2π is

$$\mathcal{F}(V,r) = \big\{ z \in \mathcal{C}(V,r) :\ 0 \le z_{1,i}^{\mathrm{f}} < 2\pi \ \ \forall i \le a \big\}.$$

Uses require $r > 0$.
Frozen Lean declaration: `KAM.Planetary.cylinder, KAM.Planetary.cylinderCell`
Source: CP10 p. 561, below (55) (disc region; audit P1); CP10 footnote 1 (period 2π).

Definition 1.11 (The family $H_0 + \mu P$)**.** For $H_0 : \mathbb{R}^a \to \mathbb{R}$ and $P : \mathbb{R} \times \mathrm{Ph}(a+b) \to \mathbb{R}$, $H_\mu(z) = H_0(z_2^{\mathrm{f}}) + \mu P(\mu, z)$.
Frozen Lean declaration: `KAM.Planetary.degenerateHamiltonian`
Source: CP10 (1), p. 546.

Definition 1.12 (Averaged perturbation)**.** For $\mu \in \mathbb{R}$, $I \in \mathbb{R}^a$, $p, q \in \mathbb{R}^b$,

$$P_{\mathrm{av}}(\mu, I; p, q) = (2\pi)^{-a} \int_{[0,2\pi]^a} P\big(\mu, \mathrm{assemble}(I, \varphi, p, q)\big)\, d\varphi$$

(Bochner integral; $= 0$ for a non-integrable integrand, so every use asserts integrability).
Frozen Lean declaration: `KAM.Planetary.secularAverage`
Source: CP10 (3), p. 547.

Definition 1.13 (Fast diffeomorphism)**.** For H_0 and $W \subset \mathbb{R}^a$, $\mathrm{HasFastDiffeo}(H_0, W)$ holds if (i) H_0 is real analytic at every point of W (`AnalyticOnNhd`); (ii) $\omega_{H_0}(W)$ is open; (iii) there is $g : \mathbb{R}^a \to \mathbb{R}^a$, of class C^1 on $\omega_{H_0}(W)$, with $g(\omega_{H_0}(I)) = I$ for $I \in W$, and $g(w) \in W$ and $\omega_{H_0}(g(w)) = w$ for $w \in \omega_{H_0}(W)$. Thus ω_{H_0} is a bijection of W onto its open image with C^1 inverse g (explicit, not a totalized inverse).
Frozen Lean declaration: `KAM.Planetary.HasFastDiffeomorphism`
Source: CP10 (A1), p. 547; neighbourhood version by audit P2.

Definition 1.14 (Secular Birkhoff form (A2"))**.** Let $V \subset \mathbb{R}^a$, $r_0, \mu_0 \in \mathbb{R}$, and functions $P_0 : \mathbb{R} \times \mathbb{R}^a \to \mathbb{R}$, $\Omega : \mathbb{R} \times \mathbb{R}^a \to \mathbb{R}^b$, $\beta : \mathbb{R} \times \mathbb{R}^a \to \mathbb{R}^{b \times b}$. $\mathrm{HasBirkhoff}(P, V, r_0, \mu_0, P_0, \Omega, \beta)$ holds if for every $\mu \in [0, \mu_0]$ and $I \in V$:

(i) for all $p, q \in \mathbb{R}^b$ with $p_j^2 + q_j^2 < r_0^2$ for all j, $\varphi \mapsto P(\mu, \mathrm{assemble}(I, \varphi, p, q))$ is integrable on $[0, 2\pi]^a$;

(ii) for every $\epsilon > 0$ there is δ with $0 < \delta \le r_0$ such that for all p, q with $p_j^2 + q_j^2 < \delta^2$ for all j,

$$\Big| P_{\mathrm{av}}(\mu, I; p, q) - \Big(P_0(\mu, I) + \sum_j \Omega_j(\mu, I) R_j + \tfrac{1}{2} \sum_{i,j} \beta_{ij}(\mu, I) R_i R_j \Big) \Big| \le \epsilon \, |(p, q)|_\infty^4,$$

where $R_j = R_j(p, q)$ and $|(p, q)|_\infty = \max(|p|_\infty, |q|_\infty)$. (The case $p = q = 0$ is included and fixes $P_{\mathrm{av}}(\mu, I; 0, 0) = P_0(\mu, I)$.)

Note that δ may depend on (μ, I, ϵ) in this definition; uniformity must be derived from the joint analyticity hypothesis of Theorem 1.28 (see `PLAN.md`, Part 5).
Frozen Lean declaration: `KAM.Planetary.HasSecularBirkhoffForm`
Source: CP10 (A2"), p. 547 (CP11 (A_2), p. 51), with the oscillator-disc guards of audit P1.

Definition 1.15 (Cylinder torus)**.** Let $H : \mathrm{Ph}(a+b) \to \mathbb{R}$, $U \subset \mathrm{Ph}(a+b)$, $w \in \mathbb{R}^{a+b}$, $K : \mathbb{R}^{a+b} \to \mathrm{Ph}(a+b)$. IsCylTorus$(H, U, w, K)$ holds if:

(i) K is real analytic on $\mathbb{R}^{a+b}$;
(ii) $K(\theta) \in U$ and H is differentiable at $K(\theta)$, for all θ;
(iii) $K(\theta + k) = K(\theta) + \mathrm{shift}\big((k_i)_{i<a}\big)$ for all $\theta \in \mathbb{R}^{a+b}$, $k \in \mathbb{Z}^{a+b}$;
(iv) if $K(\theta') = K(\theta) + \mathrm{shift}(k)$ for some $k \in \mathbb{Z}^a$, then $\theta' - \theta \in \mathbb{Z}^{a+b}$;
(v) $DK(\theta)$ is injective for all θ;
(vi) $\Omega(DK(\theta)u, DK(\theta)v) = 0$ for all θ, u, v;
(vii) for all $\theta, t \in \mathbb{R}$, $s \mapsto K(\theta + sw)$ has derivative $X_H(K(\theta + tw))$ at $s = t$.

No Diophantine condition is part of this predicate (see Definition 1.16).
Frozen Lean declaration: `KAM.Planetary.IsCylinderTorus`
Source: CP11 Thm. 11.1, pp. 50–51 (analytic conjugacy to a linear flow, Lagrangian); CP10 Thm. 1.3, p. 548.

Definition 1.16 (Cylinder Kolmogorov set)**.** For $H : \mathrm{Ph}(a+b) \to \mathbb{R}$, $V \subset \mathbb{R}^a$, $r, \mu, C \in \mathbb{R}$ and $S \subset \mathrm{Ph}(a+b)$, IsCylKol$(H, V, r, \mu, C, S)$ holds if S is Lebesgue measurable, $S \subset \mathcal{C}(V, r)$, and for every $z \in S$ there are $w \in \mathbb{R}^{a+b}$, $K : \mathbb{R}^{a+b} \to \mathrm{Ph}(a+b)$ and θ with: $K(\theta) = z$; $K(\mathbb{R}^{a+b}) \subset S$ (a whole torus); IsCylTorus$(H, \mathcal{C}(V, r), w, K)$; there are $\gamma > 0$ and $\tau > a + b$ with w (γ, τ)-Diophantine (the constant may depend on the torus); $|w^{\mathrm{f}}|_\infty \le C$ and $|w^{\mathrm{s}}|_\infty \le C\mu$.
Frozen Lean declaration: `KAM.Planetary.IsCylinderKolmogorovSet`
Source: CP11 Thm. 11.1, pp. 50–51: $\omega_1 = O(1)$, $\omega_2 = O(\mu)$, Diophantine.

1.5. **Definitions for the physical theorems.** Throughout, $G, m_0 > 0$ and the planet weights $m_i > 0$.

Definition 1.17 (Reference circular configurations and the tube)**.** For $m, a \in \mathbb{R}^n$ and $\phi \in \mathbb{R}^n$ (phases in $\mathbb{R}/\mathbb{Z}$),

$$\mathrm{circ}(\phi) = \Big(\big(a_i(\cos 2\pi\phi_i, \sin 2\pi\phi_i) \big)_i, \ \big(m_i \sqrt{G m_0 / a_i} \, (-\sin 2\pi\phi_i, \cos 2\pi\phi_i) \big)_i \Big) \in \mathrm{Ph}(2n)$$

(circular prograde Kepler configurations at coupling 0; $a_i > 0$, $G, m_0 > 0$ guard the square root and the division). For $\eta \in \mathbb{R}$ the *circular tube* is the open η-neighbourhood in the phase max norm, $U_\eta = \{ z \in \mathrm{Ph}(2n) : \exists \phi \in \mathbb{R}^n, \ |z - \mathrm{circ}(\phi)|_\infty < \eta \}$.
Frozen Lean declaration: `KAM.Planetary.circularPoint, KAM.Planetary.circularTube`
Source: Fej04r Thm. 63, p. 49 (near circular coplanar Kepler motions); ours for the explicit reference point.

Definition 1.18 (No collision, pair bounds)**.** For $q \in \mathbb{R}^{2n}$: NoColl(q) means $|q^{(i)}|_2 > 0$ for all i and $|q^{(i)} - q^{(j)}|_2 > 0$ for all $i \ne j$ (heliocentric configurations, the Sun at the origin). For N points $Q_0, \dots, Q_{N-1} \in \mathbb{R}^2$, PairBounds$(d, D; Q)$ means $d \le |Q_j - Q_i|_2 \le D$ for all $i \ne j$.
Frozen Lean declaration: `KAM.Planetary.NoCollision, KAM.Planetary.PairBounds`
Source: Fej04r §6.1, pp. 45–47.

Definition 1.19 (Actual masses, barycentric reconstruction)**.** For $\mu \in \mathbb{R}$ let $\mathrm{mass}_\mu = (m_0, \mu m_1, \dots, \mu m_n) \in \mathbb{R}^{n+1}$ (index 0 is the Sun). For heliocentric positions $q \in \mathbb{R}^{2n}$ and scaled momenta $p \in \mathbb{R}^{2n}$,

$$x_0(q) = -\frac{\mu}{m_0 + \mu \sum_i m_i} \sum_i m_i q^{(i)}, \qquad x_i(q) = x_0(q) + q^{(i)},$$

$$v_0(p) = -\frac{\mu}{m_0} \sum_i p^{(i)}, \qquad v_i(p) = \frac{1}{m_i} p^{(i)} \qquad (1 \le i \le n),$$

with the Sun at index 0 and planet i at index i in the order of `Fin.cons`. The denominators are positive for $m_i > 0$, $\mu > 0$.
Frozen Lean declaration: `KAM.Planetary.actualMasses, KAM.Planetary.barycentricPositions, KAM.Planetary.barycentricVelocities`
Source: Fej04r §6.1, (28), pp. 46–47; zero total momentum and barycentre; Fej07e p. 3.

Definition 1.20 (Newton solutions)**.** Let $N \in \mathbb{N}$, $M \in \mathbb{R}^N$ and $Q, V : \mathbb{R} \to (\mathbb{R}^2)^N$. $\mathrm{IsNewton}(G, M, Q, V)$ holds if for all $t \in \mathbb{R}$ and $i \neq j$ (all indices $< N$): (i) $|Q_j(t) - Q_i(t)|_2 > 0$; (ii) Q_i has derivative $V_i(t)$ at t; (iii) V_i has derivative

$$\sum_{j \neq i} \frac{G M_j}{|Q_j(t) - Q_i(t)|_2^3} \, (Q_j(t) - Q_i(t)) \quad \text{at } t$$

(the all-time Newton equations; `HasDerivAt`).
Frozen Lean declaration: `KAM.Planetary.IsNewtonSolution`
Source: Fej04r (1), §6.1, with G restored.

Definition 1.21 (Fixed-Sun solutions)**.** For $Q, V : \mathbb{R} \to (\mathbb{R}^2)^n$ and $\varepsilon \in \mathbb{R}$, the predicate $\mathrm{IsFixedSol}(G, m_0, m, \varepsilon, Q, V)$ holds if for all t and $i \neq j$: $|Q_i(t)|_2 > 0$, $|Q_j(t) - Q_i(t)|_2 > 0$, Q_i has derivative $V_i(t)$ and V_i has derivative

$$-\frac{G m_0}{|Q_i(t)|_2^3} \, Q_i(t) + \varepsilon \sum_{j \neq i} \frac{G m_j}{|Q_j(t) - Q_i(t)|_2^3} (Q_j(t) - Q_i(t)).$$

The Sun is the constant origin.
Frozen Lean declaration: `KAM.Planetary.IsFixedSolution`
Source: Ours (fixed-center law obtained by differentiating H_{fix}), after Fej04r (28).

Definition 1.22 (Moving-Sun Hamiltonian)**.** For $z = (q, p) \in \mathrm{Ph}(2n)$ and $\mu \in \mathbb{R}$,

$$H_{\mathrm{mov}}(z) = \sum_i \Big[\big(\tfrac{1}{m_i} + \tfrac{\mu}{m_0}\big) \tfrac{|p^{(i)}|_2^2}{2} - \frac{G m_0 m_i}{|q^{(i)}|_2} \Big] + \mu \sum_{i<j} \Big[\frac{p^{(i)} \cdot p^{(j)}}{m_0} - \frac{G m_i m_j}{|q^{(i)} - q^{(j)}|_2} \Big].$$

It is the reduced Hamiltonian divided by μ, with G restored, and every term is kept: the Sun moves, and neither the self kinetic term $\mu |p^{(i)}|_2^2/(2m_0)$ nor the cross term is dropped. Equivalently $H_{\mathrm{mov}} = H_{\mathrm{fix}}\big|_{\varepsilon=\mu} + \mu |\sum_i p^{(i)}|_2^2/(2m_0)$ (a calculation for a later part).
Frozen Lean declaration: `KAM.Planetary.movingHamiltonian`
Source: Fej04r (28), pp. 46–47, with Fej07e p. 3 (indices $m_j m_k$ corrected) and G restored.

Definition 1.23 (Fixed-Sun Hamiltonian)**.** For $z = (q, p) \in \mathrm{Ph}(2n)$ and $\varepsilon \in \mathbb{R}$,

$$H_{\mathrm{fix}}(z) = \sum_i \Big[\frac{|p^{(i)}|_2^2}{2 m_i} - \frac{G m_0 m_i}{|q^{(i)}|_2} \Big] - \varepsilon \sum_{i<j} \frac{G m_i m_j}{|q^{(i)} - q^{(j)}|_2}.$$

The Sun is the origin; only the direct planet–planet interaction is kept, with coefficient ε.
Frozen Lean declaration: `KAM.Planetary.fixedHamiltonian`
Source: Ours (Newtonian fixed-center model; "our variant of Fej04r (28)"); `statement.md`, P-FIXED.

Definition 1.24 (Physical torus)**.** For $H : \mathrm{Ph}(d) \to \mathbb{R}$, $w \in \mathbb{R}^d$ and $K : \mathbb{R}^d \to \mathrm{Ph}(d)$, $\mathrm{IsPhysTorus}(H, w, K)$ holds if:

(i) K is real analytic on $\mathbb{R}^d$;
(ii) H is differentiable at $K(\theta)$ for all θ;
(iii) $K(\theta + k) = K(\theta)$ for $k \in \mathbb{Z}^d$;
(iv) $K(\theta) = K(\theta')$ implies $\theta' - \theta \in \mathbb{Z}^d$;
(v) $DK(\theta)$ is injective;
(vi) $\Omega(DK(\theta)u, DK(\theta)v) = 0$;
(vii) there are $\gamma > 0$ and $\tau > d - 1$ with w (γ, τ)-Diophantine;
(viii) for all θ, t, $s \mapsto K(\theta + sw)$ has derivative $X_H(K(\theta + tw))$ at $s = t$.

(For such K the image is a compact embedded analytic Lagrangian torus. Track K's primary-torus predicate has affine winding and is not reused. Note $\tau > d - 1$ here against $\tau > a + b$ in Definition 1.16.)
Frozen Lean declaration: `KAM.Planetary.IsPhysicalTorus`
Source: HL19 (2.1), (2.3), pp. 6–7 (modified to period-one embeddings); CP09 Thm. 4, p. 8; ours.

Definition 1.25 (Moving-Sun Kolmogorov set)**.** For $G, m_0 > 0$, $m, a \in \mathbb{R}^n$, $\mu, \eta, d, D \in \mathbb{R}$, $S \subset \mathrm{Ph}(2n)$: $\mathrm{MovKol}(G, m_0, m, a, \mu, \eta, d, D, S)$ holds if S is measurable, $S \subset U_\eta$, and every $z \in S$ has w, K, θ with $K(\theta) = z$, $K(\mathbb{R}^{2n}) \subset S$, $\mathrm{IsPhysTorus}(H_{\mathrm{mov}}, w, K)$ and, for every $\phi \in \mathbb{R}^{2n}$:

(a) $\mathrm{NoColl}\big(K(\phi)_1\big)$;
(b) $\mathrm{IsNewton}\big(G, \mathrm{mass}_\mu,\ t \mapsto x(K(\phi + tw)_1),\ t \mapsto v(K(\phi + tw)_2)\big)$, where x, v are the barycentric reconstructions of Definition 1.19;
(c) $\mathrm{PairBounds}\big(d, D;\ x(K(\phi)_1)\big)$ (all pairs including the Sun; as ϕ is arbitrary this covers every point $K(\phi + tw)$ of every orbit, hence all times).

Frozen Lean declaration: `KAM.Planetary.MovingKolmogorovSet`
Source: Fej04r Prop. 77 and Fej07e §7, pp. 3–4, with CP09 Thm. 4; the physical conclusions are ours (`statement.md`).

Definition 1.26 (Fixed-Sun Kolmogorov set)**.** For ε, η, d, D and $S \subset \mathrm{Ph}(2n)$ the predicate $\mathrm{FixKol}(G, m_0, m, a, \varepsilon, \eta, d, D, S)$ holds if S is measurable, $S \subset U_\eta$, and every $z \in S$ has w, K, θ with $K(\theta) = z$, $K(\mathbb{R}^{2n}) \subset S$, $\mathrm{IsPhysTorus}(H_{\mathrm{fix}}, w, K)$ and for every ϕ: $\mathrm{NoColl}(K(\phi)_1)$; $\mathrm{IsFixedSol}\big(G, m_0, m, \varepsilon,\ t \mapsto (K(\phi + tw)_1^{(i)})_i,\ t \mapsto (\frac{1}{m_i} K(\phi + tw)_2^{(i)})_i\big)$; $\mathrm{PairBounds}\big(d, D;\ (0, K(\phi)_1^{(1)}, \dots, K(\phi)_1^{(n)})\big)$, with the Sun at the origin as point 0.
Frozen Lean declaration: `KAM.Planetary.FixedKolmogorovSet`
Source: Ours (new deduction; Fej04r Lemma 64 and Prop. 77, CP09 Thm. 4).

Remark 1.27 (Lean totality). The frozen definitions use Lean total functions, and the frozen theorems guard each of them as follows; proofs must establish these guarded facts before using the defining formulas.

- *Fréchet derivatives* (ω_{H_0}, X_H, DK) are 0 at points of non-differentiability. ω_{H_0} is used only through `HasFastDiffeomorphism`, which asserts analyticity of H_0 on $W \supset \overline{V}$; X_H appears only in the clauses (vii)/(viii) of the torus predicates, which also assert differentiability of H on the range of K (for the cylinder tori at points of $\mathcal{C}(V, r)$, where H_μ is analytic because $r < r_0$ and $\mu \in [0, \mu_0]$, by the hypotheses on P and H_0); DK is guarded by analyticity of K. Injectivity of DK and $\Omega(DK\cdot, DK\cdot) = 0$ are asserted for the genuine derivative.

- *Bochner integrals* (P_{av}) are 0 for non-integrable integrands; Definition 1.14 (i) asserts integrability on the disc $p_j^2+q_j^2<r_0^2$, and every point used in (ii) has $\delta\le r_0$.
- *Square roots and divisions*: $\sqrt{\cdot}$ in $|\cdot|_2$ has nonnegative radicand; $1/|\cdot|_2$ and $1/|\cdot|_2^3$ occur only after the collision-free clauses (a), IsNewton (i), IsFixedSol (the first two conjuncts) in the same conjunction; in circ, $\sqrt{Gm_0/a_i}$ and $1/a_i$ are guarded by $G,m_0,a_i>0$; in x_0 and v_0 the denominators $m_0+\mu\sum m_i$ and m_0 are positive since $\mu>0$; the factor $1/m_i$ needs $m_i>0$.
- *Real powers and logarithms*: $r^6/(\log(1/r))^c$ with $0<r<r_*<1$ has positive base $\log(1/r)>0$; $\sqrt{r}$ has $r>0$; $|k|_1^\tau$ in the Diophantine condition has $|k|_1\ge 1$ for $k\neq 0$.
- *Truncated subtraction*: the lower bound is written with `ENNReal.ofReal`, so it is the real number $1-C\sqrt{r}$ when that is positive and 0 otherwise; the strict inequality $\mathrm{ofReal}(1-C\sqrt{r})\,\mathrm{meas}\,\mathcal{F}<\mathrm{meas}(S\cap\mathcal{F})$ is then nontrivial only when $1-C\sqrt{r}>0$, but must be proved for every $r<r_*$ as stated (it follows from $0<\mathrm{meas}(S\cap\mathcal{F})$ when $1-C\sqrt{r}\le 0$).
- *Measure*: meas is Lebesgue (outer) measure; the sets $\mathcal{F}(V,r)$ are Borel and S is measurable by the conclusion; $\mathrm{meas}\,\mathcal{F}(V,r)<\infty$ since V is bounded.

1.6. The frozen theorems.

Theorem 1.28 (P-G: corrected properly degenerate KAM)**.** ***Data (all fixed first).*** *Integers $a,b\ge 1$. A set $V\subset\mathbb{R}^a$ that is open, connected and bounded; an open set $W\subset\mathbb{R}^a$ with $\overline{V}\subset W$; a boundary collar condition: there are $A,d>0$ such that for all $0<t<d$*

$$\mathrm{meas}\{I\in\mathbb{R}^a:\ \exists J\in\partial V,\ |I-J|_\infty<t\}\le A\,t.$$

Reals $r_0,\mu_0,B>0$. Functions $H_0:\mathbb{R}^a\to\mathbb{R}$, $P:\mathbb{R}\times\mathrm{Ph}(a+b)\to\mathbb{R}$, $P_0:\mathbb{R}\times\mathbb{R}^a\to\mathbb{R}$, $\Omega:\mathbb{R}\times\mathbb{R}^a\to\mathbb{R}^b$, $\beta:\mathbb{R}\times\mathbb{R}^a\to\mathbb{R}^{b\times b}$.

Hypotheses.

(H1) *$(\mu,z)\mapsto P(\mu,z)$ is real analytic at every point of $[0,\mu_0]\times\overline{\mathcal{C}(V,r_0)}$ (jointly, as a function on $\mathbb{R}\times\mathrm{Ph}(a+b)$);*
(H2) *for $\mu\in[0,\mu_0]$, $z\in\mathcal{C}(V,r_0)$ and $k\in\mathbb{Z}^a$: $P(\mu,z+\mathrm{shift}(k))=P(\mu,z)$;*
(H3) $\mathrm{HasFastDiffeo}(H_0,W)$ *(Definition 1.13);*
(H4) $\mathrm{HasBirkhoff}(P,V,r_0,\mu_0,P_0,\Omega,\beta)$ *(Definition 1.14);*
(H5) *$|\det\beta(\mu,I)|\ge B$ for all $\mu\in[0,\mu_0]$, $I\in V$;*
(H6) *$\beta(\mu,I)$ is symmetric for all $\mu\in[0,\mu_0]$, $I\in V$.*

Conclusion. *There exist r_*,C,c with $0<r_*<r_0$, $r_*<1$, $C>0$, $c>0$ such that for all r,μ with*

$$0<r<r_*,\qquad 0<\mu<\mu_0,\qquad \mu<\frac{r^6}{(\log(1/r))^c},$$

there is $S\subset\mathrm{Ph}(a+b)$ with $\mathrm{IsCylKol}(H_\mu,V,r,\mu,C,S)$ *(with $H_\mu=H_0+\mu P(\mu,\cdot)$ of Definition 1.11) and*

$$0<\mathrm{meas}(S\cap\mathcal{F}(V,r)),\qquad \mathrm{ofReal}(1-C\sqrt{r})\,\mathrm{meas}\,\mathcal{F}(V,r)<\mathrm{meas}(S\cap\mathcal{F}(V,r))<\mathrm{meas}\,\mathcal{F}(V,r).$$

Here r_*,C,c depend on all the data and hypotheses above and not on r or μ; C is simultaneously the frequency bound of Definition 1.16 and the measure constant.

Frozen Lean declaration: `KAM.Planetary.properlyDegenerateKAM`

Source: CP11 Thm. 11.1, pp. 50–51, whose proof is CP10 Thm. 1.3, p. 548 (with $c_*=2b$, $\rho=r$). It is not *the printed theorem: the neighbourhood W (CP10 p. 556, (58) p. 562), the collar (CP10 p. 561, below (56)), the product of discs (CP10 p. 561, below (55)), joint analyticity of the family (CP10 footnote 3, p. 546 asks only C^1 uniform dependence) and the 2π period are corrected-contract premises (audit P1–P4; D-PG-1 to D-PG-5).*

Source gap: CP11 Thm. 11.1 states $\mathrm{meas}\,P_\epsilon>\mathrm{meas}\,K$ *without proof; the strict inequality* $\mathrm{meas}(S\cap\mathcal{F})<\mathrm{meas}\,\mathcal{F}$ *is obtained in the plan by deleting a small positive-measure subfamily of tori (Part 5), which is*

possible because the lower bound has slack. CP10 Thm. 1.3 also needs $P_{\mathrm{av}} - N = O(\epsilon^5)$ uniformly (CP10 (60), p. 562), which follows from (H1), (H2), (H4) and compactness but is not proved in the source.

Theorem 1.29 (P-PLANAR: two planar planets with a moving Sun)**.** *Let $G, m_0 > 0$; let $m, a : \mathit{Fin}\,2 \to \mathbb{R}$ (planets $0, 1$; m the planet weights, a the axes) with $m_i > 0$ and $a_i > 0$ for $i = 0, 1$ and $a_0 < a_1$ (strictly ordered axes, no numerical separation); let $\eta > 0$. Then there exist μ_0, d, D with $0 < \mu_0$ and $0 < d < D$ such that for every μ with $0 < \mu < \mu_0$ there is $S \subset \mathrm{Ph}(2 \cdot 2)$ with*

$$0 < \operatorname{meas} S < \infty \quad \textit{and} \quad \mathrm{MovKol}(G, m_0, m, a, \mu, \eta, d, D, S) \qquad (n = 2).$$

The order is: data (G, m_0, m, a, η), then (μ_0, d, D), then μ, then S. The tori are 4-dimensional in the 8-dimensional $\mathrm{Ph}(4)$.

Frozen Lean declaration: `KAM.Planetary.planarTwoPlanetKAM`

Source: CP09 Thm. 4, p. 8 with Fej04r Prop. 77, pp. 56–58, using the recorded source-proof repairs (`statement.md`, P5–P7); Fej07e p. 3–4 supplies the smooth counterpart. No threshold from CP11 §8.1 (audit P4).

Source gap: The route is not signed off as proved: the analytic weak-nondegeneracy gate (CP09 Thm. 4 resting on Rüssmann's theorem, CP09 Thm. 3, whose source is not pinned), the continuation of eigenvalue branches (Fej04r Lemma 76 is false as printed) and the closeness of the tori to the circular torus (CP09 Thm. 4 says only "close to"); see `PLAN.md`.

Theorem 1.30 (P-FIXED: n planar planets around a fixed Sun)**.** *Let $n \geq 2$, $G, m_0 > 0$, $m, a \in \mathbb{R}^n$ with all $m_i > 0$ and $a_i > 0$, $a_i < a_j$ whenever $i < j$, and $\eta > 0$. Then there exist ε_0, d, D with $0 < \varepsilon_0$ and $0 < d < D$ such that for every ε with $0 < \varepsilon < \varepsilon_0$ there is $S \subset \mathrm{Ph}(n \cdot 2)$ with*

$$0 < \operatorname{meas} S < \infty \quad \textit{and} \quad \mathrm{FixKol}(G, m_0, m, a, \varepsilon, \eta, d, D, S).$$

The order is data (n, G, m_0, m, a, η), then (ε_0, d, D), then ε, then S.

Frozen Lean declaration: `KAM.Planetary.fixedSunPlanetaryKAM`

Source: Ours: "our proposed Newtonian fixed-center deduction, not a published fixed-Sun statement"; Fej04r Lemma 64 (p. 49–50) and Prop. 77, CP09 Thm. 4.

Source gap: The secular identification for the fixed Sun (equal secular quadratic forms and frequency map up to functions of the actions) and the general-n induction of Fej04r Prop. 77 ("par un raisonnement analogue", p. 58) are proof obligations; no secular nondegeneracy is a premise.

Remark 1.31 (Scope of this part). The three statements are the only frozen theorems of track P. P-SPATIAL is a sketch only and has no Lean statement. The D22 additions are under statement audit and are treated in the reserved last part.